University Comparison Visualizer

About the Visualization

I am interested in finding how universities compare to one another in terms of performance. There are many online publications of university rankings, such as by *Quacquarelli Symonds* (QS) and *Times Higher Education* (THE). However, I find that these websites do not allow much freedom for visitors to compare between universities. For instance, QS limits comparisons to four universities at a time and only displays a general summary table of each university on its comparisons page.

Therefore, I have developed a Shiny application that visualizes university performance and allows users to make comparisons between them and based on several metrics such as region, research output, and type. This application answers:

1. How do universities compare by region?
2. What is the score gap between public and private universities?
3. Does research output impact university score?
4. How does a university stand out from its neighbors over the years?

I believe my application can benefit students and parents in having a bigger picture of each institution and also university administrators in formulating ideas to improve performance.

Interesting Findings

My visualizations revealed several fascinating insights. The most unexpected finding would be that the average university scores across all regions declined from 2017 to 2022. Surprisingly, the most significant decline was in 2019 before COVID-19. The trend for subsequent years remained relatively flat, except for African universities. My best reasoning for this is that there was a major overhaul of the scoring methodology. Certain criteria which many universities underperform in might have been weighted more heavily. As a consequence, their scores were negatively affected.

In addition, the score gap between private and public universities are getting closer each year. This disproves the notion that private institutions are superior compared to their public counterparts. Such scenario can be attributed to the fact that many public universities might have improved research output and adopted strategies for boosting education quality.

Interface Creation Process

Kaggle's *QS World University Rankings 2017-2022* dataset that I used for this interface required several preparations. I reordered the factor levels for the university size and research output variables accordingly. Moreover, I replaced the NA values with 0 for the columns related to university population to ease filtering operations. Since certain university ranks are in the form of range, I have also created a helper function to calculate the mean of the upper and lower limit, consequently converting the column into a numeric data type. This enabled me to plot university rankings over time.

I designed the interface with the idea of it being coherent, user-friendly, and complete with useful information. I have made multiple display modes, "General" and "Specific", controlled by radio buttons to suit user needs. The "General" mode displays average university scores based on region, type, and research output. I wanted users to have the ability to customize the plots; hence, I added three options to filter by specific ranges of university population. Furthermore, I used the five colors of the Olympic rings to represent each region for recognizability and added a purple color for Latin America, while I standardized the colors of the remaining "General" plots for simplicity.

The "Specific" mode shows the historical performance of a chosen university. To streamline interaction, I made the rankings table clickable so users can select the focus university without the need for a dropdown menu. Taking a step further, I have added the ability for users to visualize the performance of multiple universities on a single graph. I have also color-coded the bump chart, displaying universities with lower average performance relative to the focus university in red and vice versa. This customization aids users in understanding how each university compares to one another. Since certain universities do not have complete data, I included links to their respective webpages so that users can still learn about their historical performances.

Reactive Graph

Considering the many features of this application, its reactive graph structure will naturally be involved. To summarize, most input are fed into the current_data reactive expression as this will be the main data for creating graphs. Additionally, the filtered_uni_data reactive expression contains data pertaining to a specific university. Focusing on the output, dynamic_ui updates the user interface according to the chosen mode, while avg_score and uni_rank_score create complex plots. Hence, they are linked to more input and reactive expressions compared to the other outputs.

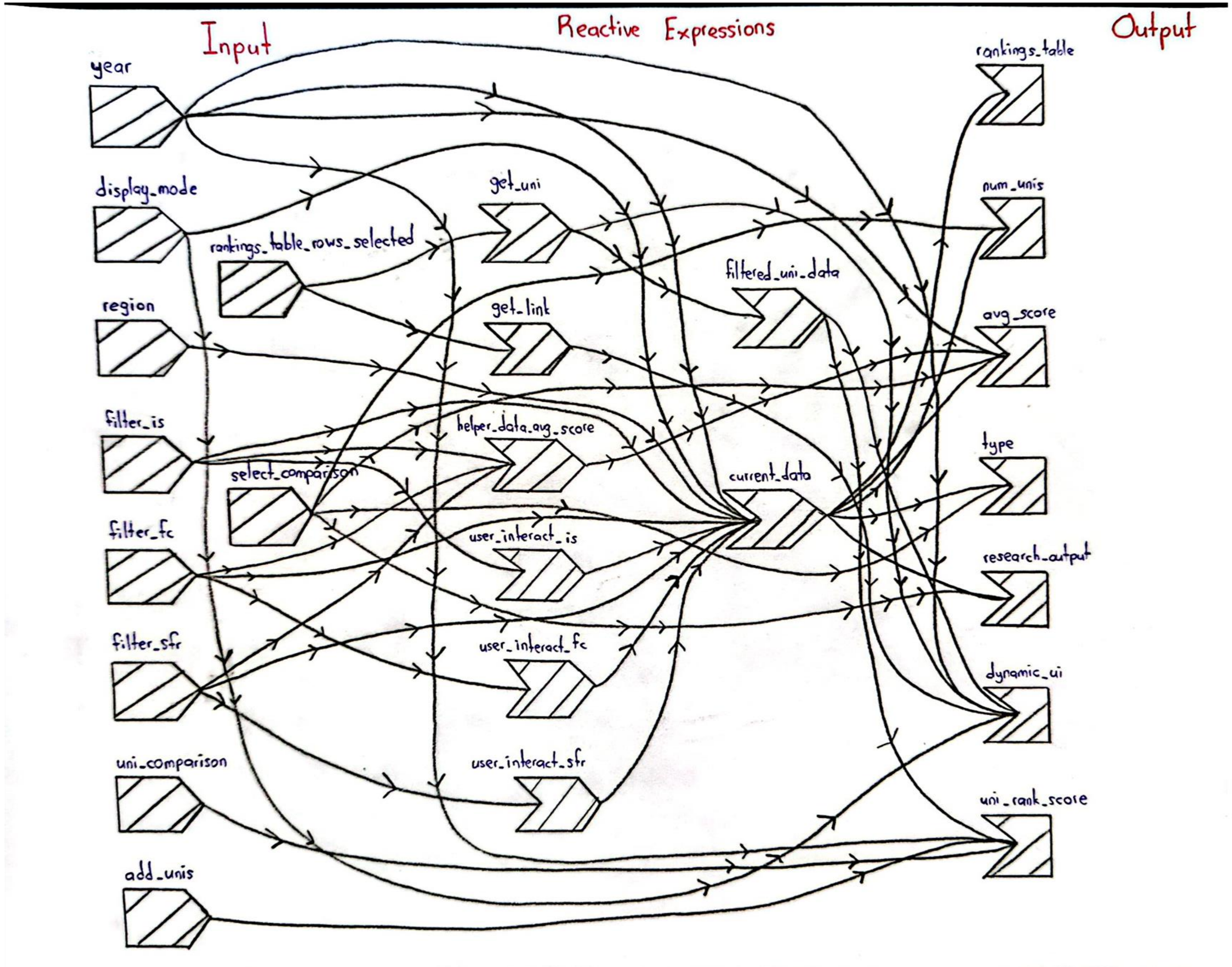

Figure 1 Reactive Graph of the Application.